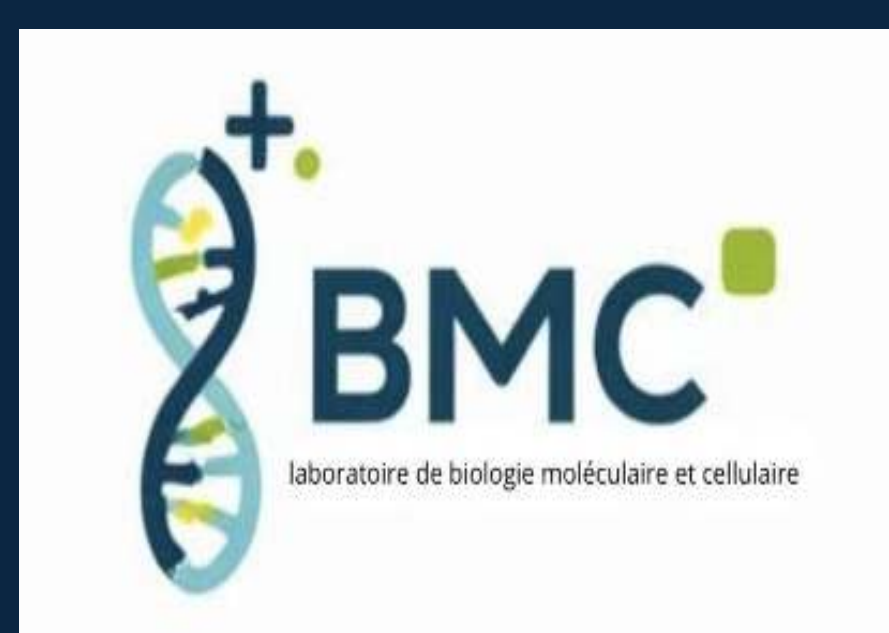




Therapeutic Patterns and Glycemic Control in Type 2 Diabetes: A Cross-Sectional Study in a Public Health Setting in Guelma, Algeria

W. ABDAOUI ^{1,3*}, I. MOKHTARI⁴, A. BRAIK ^{2,5}, D.E. BENOURETH ^{2,3}, N. HAFERSSAS², H. BENAYACHE ²

¹Department of Natural and Life Sciences, University 8th May 1945, Algeria

²Department of Biology, University 8th May 1945, Algeria

³Molecular and Cellular Biology Laboratory, University 8th May 1945, Algeria

⁴Département de pharmacie, Faculté de médecine, Université de Tlemcen

⁵ Laboratoire de chimie physique University 8th May 1945, Algeria

Introduction

- Type 2 diabetes is the most common form of the disease and most often affects adults, although an increasing number of children and adolescents.
- In people with type 2 diabetes, the body is able to produce insulin, but either the amount produced is insufficient, or the body does not respond to the hormone as it should because the cells are resistant or insensitive to insulin.
- Oral treatments for type 2 diabetes have become much more diverse over the past decade, with the introduction of glitazones, followed by gliptins, and, most recently, gliflozins.
- However, the two oral antidiabetic drugs developed more than 50 years ago—metformin and sulfonylureas—remain the market leaders.

Methodology

Study Design

This is a descriptive, cross-sectional study involving a sample of 150 patients diagnosed with type 2 diabetes (103 women and 47 men) who are receiving follow-up care at the “Oumeddour Tounes” public health facility in the Guelma region

Sampling

This is a random sample. All adult patients with type 2 diabetes of various ages who meet the inclusion criteria and agree to participate in the study are included in this study.

Study Period

It was conducted over a period of two months. Data collection began during an internship from February 4, 2024, through April 4, 2024.

Inclusion criteria

All consecutive patients with type 2 diabetes mellitus, of both sexes, aged 18 years or older, residing in Guelma, and presenting either to the emergency department or for routine follow-up at the diabetology unit, were included. The diagnosis of type 2 diabetes had to be confirmed by the attending physician. Only those who agreed to complete the questionnaire and provided a written informed consent were enrolled in the study.

Goals of the study

The primary objective of this study is to describe the various medications used in the treatment of type 2 diabetes and to evaluate their effectiveness based on achievement of the HbA1c target.

The secondary objectives of this study are:

1.To describe the epidemiological, clinical, and biological characteristics of 150 patients with type 2 diabetes followed at the “Oumeddour Tounes” public health facility in the Guelma region.

2.To present the various medications used in the treatment of type 2 diabetes as monotherapy, dual therapy, or triple therapy.

Results and Discussion

- The findings indicate a marked female predominance among the study population, with women accounting for 68.67% of cases versus 31.33% for men.
- The most frequently represented age group was [40–59] years, comprising 49.33% of the cohort.
- Anthropometric assessment revealed that all patients had a body mass index (BMI) ≥ 25 kg/m², with overweight observed in 52.66% and obesity in 36.66% of the cases.
- Several factors were found to be associated with the onset of type 2 diabetes mellitus, including advancing age, positive family history of diabetes, arterial hypertension, sedentary lifestyle, tobacco use, and unbalanced dietary practices.
- Regarding pharmacological management, metformin was the most commonly prescribed agent, used in 83.33% of patients.
- Furthermore, a comparison between current and previous HbA1c levels revealed a statistically significant difference, suggesting an improvement or change in glycemic control over time
- Our major finding is the significant improvement in mean HbA1c (a reduction of 0.37%). Although modest, this reduction is clinically meaningful. In the landmark UKPDS trial, every 1% reduction in HbA1c was associated with a 21% decrease in diabetes-related mortality [5]. However, with a current mean HbA1c of 7.78%, nearly half (43.6%) of our patients remain above the ADA-recommended target of <7% [3], exposing them to a persistent residual risk of complications.

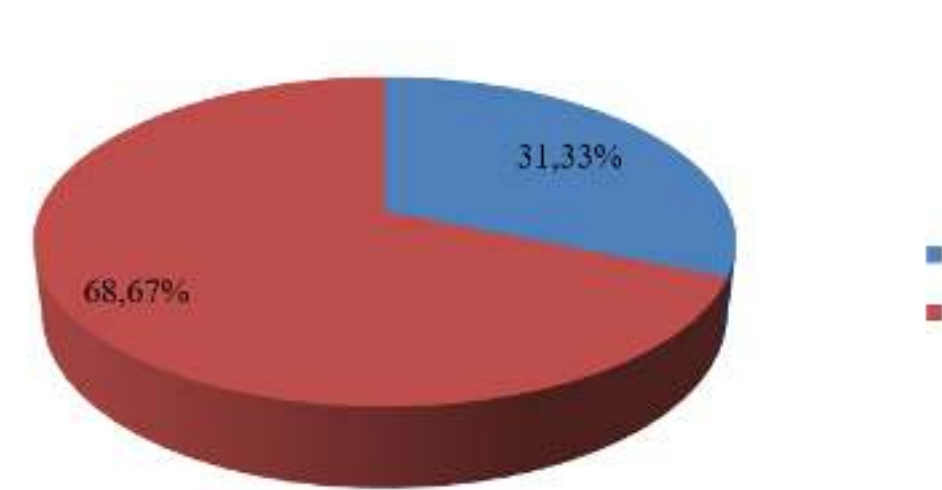
Tableau 1 – Caractéristiques générales de la population (N=150)

Variable	Moyenne $\pm$ Écart-type ou n (%)
Âge (ans)	59,5 $\pm$ 10,8
Sexe féminin	103 (68,7 %)
IMC (kg/m ²)	29,9 $\pm$ 4,2
Catégorie IMC :	
- Normal (<25)	18 (12,0 %)
- Surpoids (25-29,9)	51 (34,0 %)
- Obésité modérée (30-34,9)	57 (38,0 %)
- Obésité sévère (≥ 35)	24 (16,0 %)
PAS (mmHg)	12,6 $\pm$ 1,8
PAD (mmHg)	7,6 $\pm$ 1,1
Tabagisme actif	6 (4,0 %)
Activité physique régulière	82 (54,7 %)
Antécédent HTA	87 (58,0 %)
Antécédent familial DT2	108 (72,0 %)
HbA1c précédente (%)	8,15 $\pm$ 1,72
HbA1c actuelle (%)	7,78 $\pm$ 1,58
Δ HbA1c (actuelle – précédente)	-0,37 $\pm$ 1,45
Nombre de traitements :	
- Monothérapie	42 (28,0 %)
- Bithérapie	78 (52,0 %)
- Trithérapie	30 (20,0 %)
Sous insuline	52 (34,7 %)
Sous metformine	128 (85,3 %)

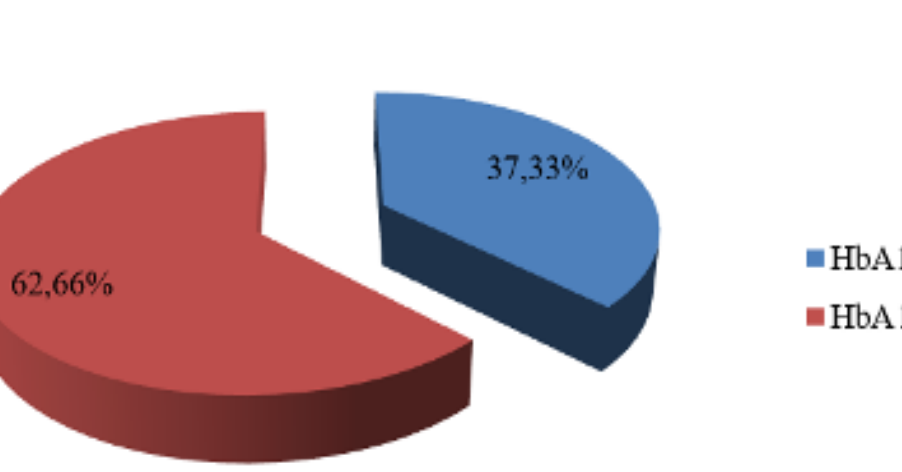
Table 2: Comparison of current and previous HbA1C mean values using the Student's t-test

	Moyenn e	Écart-type	Différence moyenne	IC 95%	p-value
HbA1c précédente	8,15 %	1,72	-0,37	[-0,59 ; -0,15]	0,002
HbA1c actuelle	7,78 %	1,58			

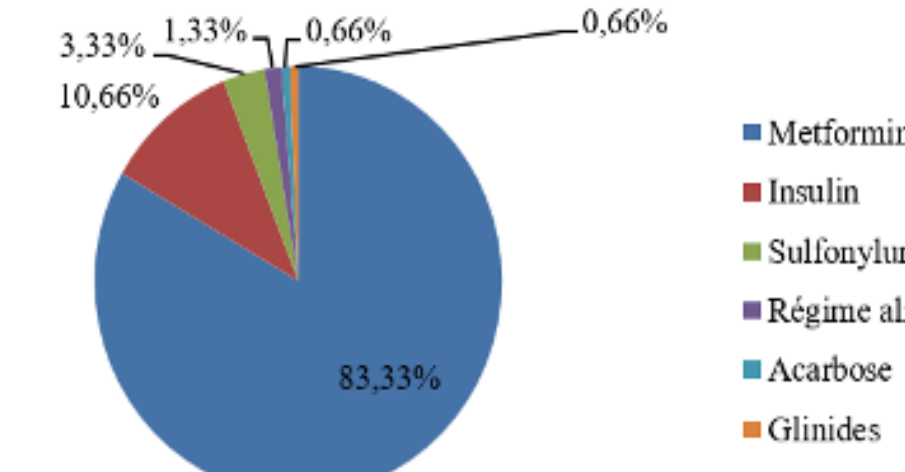
Répartition des patients en fonction du sexe



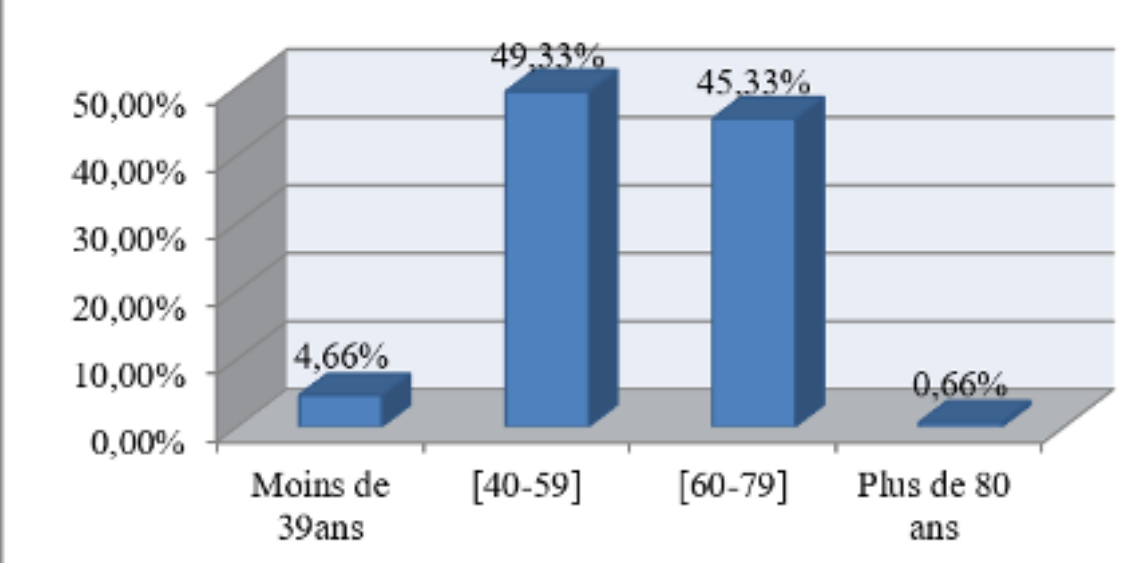
Répartition des patients en fonction de l'objectif glycémique atteint



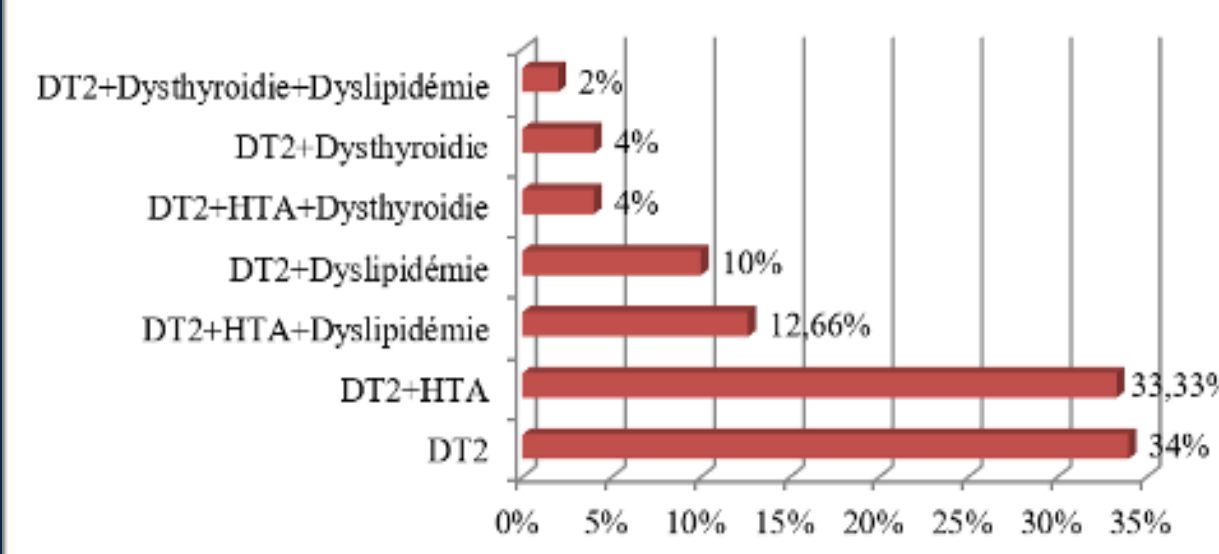
Répartition des patients en fonction du traitement antidiabétique



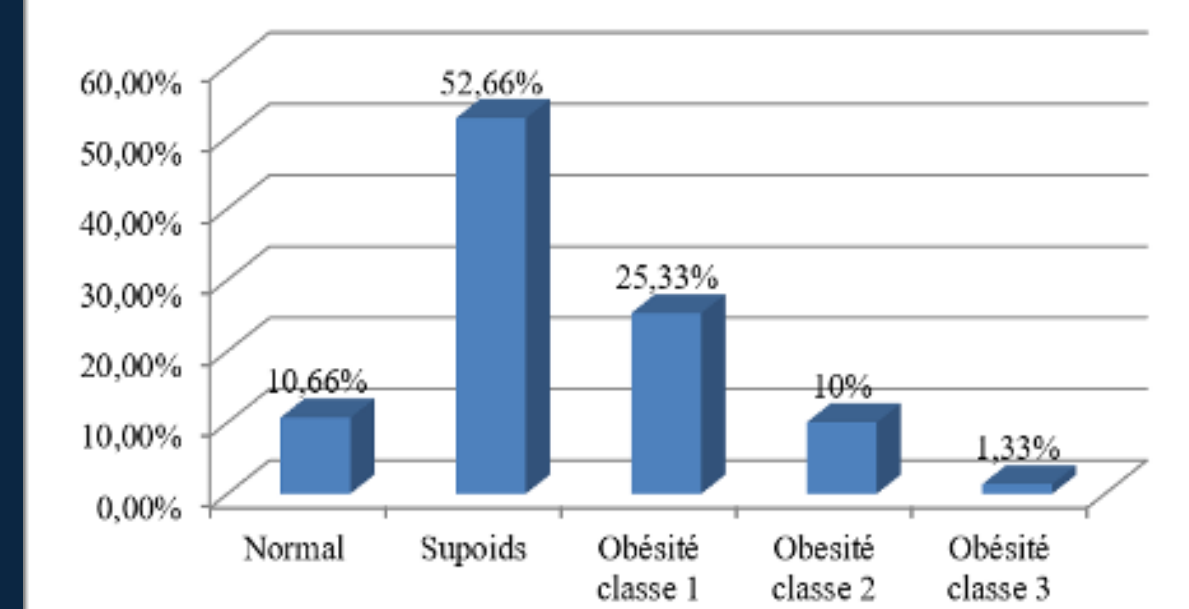
Répartition des patients en fonction de l'âge



Répartition des patients en fonction des antécédents médicaux peronnels

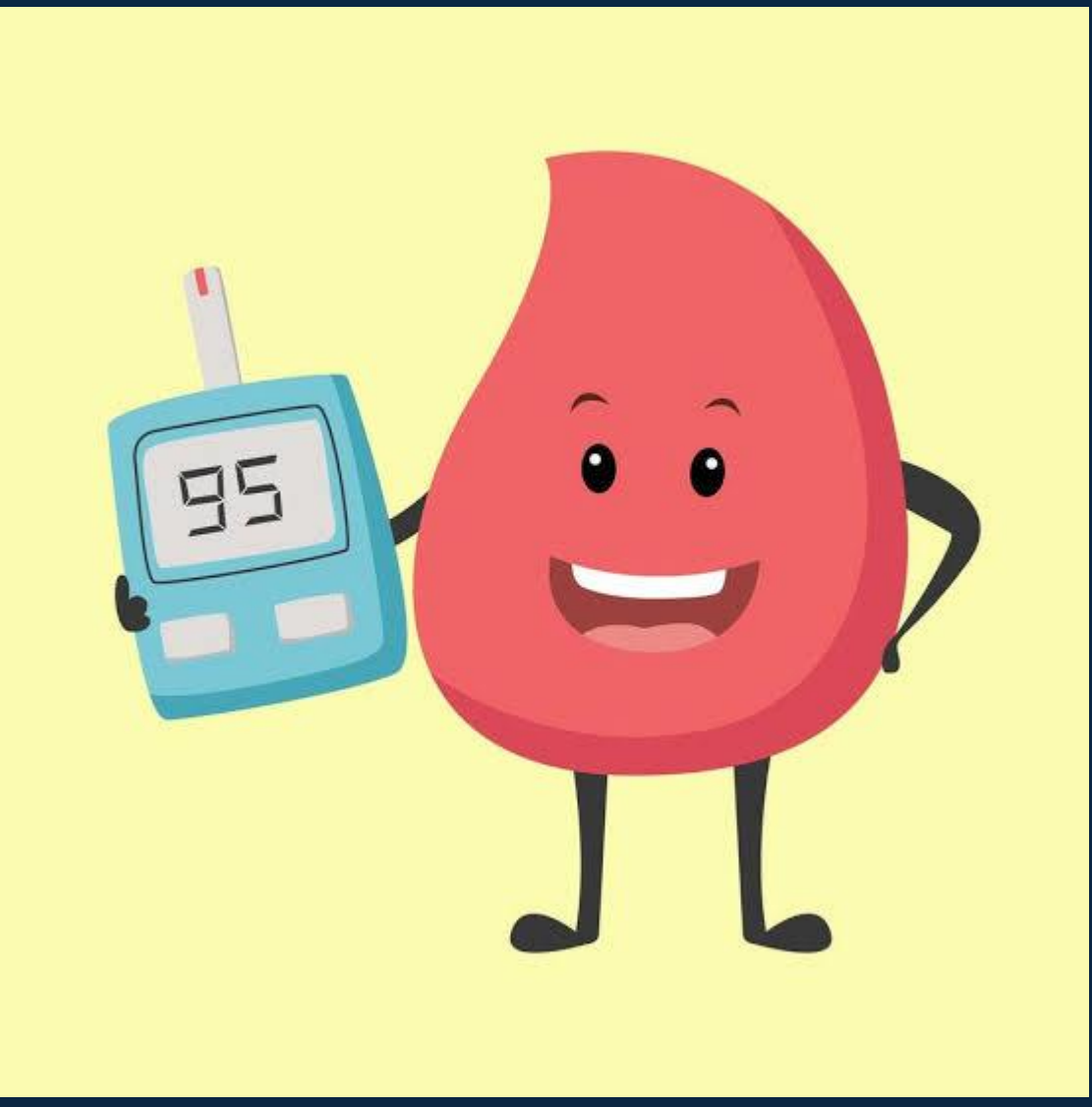


Répartition des patients selon l'IMC



Conclusion

- This study confirms that type 2 diabetes in the Guelma region predominantly affects middle-aged women and is strongly linked to modifiable risk factors such as overweight, obesity, and sedentary behaviour.
- Metformin was the most frequently prescribed antidiabetic agent. The significant improvement in HbA1c levels over time indicates satisfactory glycaemic control under current therapeutic strategies.
- Nevertheless, enhanced screening programmes and lifestyle interventions are warranted to reduce disease burden and prevent long-term complications.



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